

TCP协议分析

分析各报文段的作用

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> Frame 2177: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{8A2BD733-6604-40FC-8041-F4CFBABB1E1E}, id 0
> Ethernet II, Src: Intel_70:f5:94 (4c:77:cb:70:f5:94), Dst: zte_c1:2a:f3 (78:90:a2:c1:2a:f3)
> Internet Protocol Version 4, Src: 192.168.1.17, Dst: 120.92.211.176
> Transmission Control Protocol, Src Port: 1483, Dst Port: 23, Seq: 0, Len: 0
  Source Port: 1483
  Destination Port: 23
  [Stream index: 57]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 3834764008
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x002 (SYN)
  Window: 64240
  [Calculated window size: 64240]
  Checksum: 0xa0fc [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted
  > [Timestamps]
```

source port:源端口

Destination port:目的端口

Sequence number:序号

Acknowledgement number:确认序号

Header Length:头部长度

Flags:标志位

Widow size value:窗口大小

Checksum:校验和

Urgent pointer:紧急指针

时序关系

建立连接

ip.addr == 120.92.211.176						
No.	Time	Source	Destination	Protocol	Length	Info
2177	22.360181	192.168.1.17	120.92.211.176	TCP	66	1483 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2184	22.369393	120.92.211.176	192.168.1.17	TCP	66	23 → 1483 [SYN, ACK] Seq=0 Ack=1 Win=14400 Len=0 MSS=1440 SACK_PERM WS=512
2185	22.369482	192.168.1.17	120.92.211.176	TCP	54	1483 → 23 [ACK] Seq=1 Ack=1 Win=263424 Len=0
2186	22.381089	120.92.211.176	192.168.1.17	TELNET	57	Telnet Data ...

1.主机发起一个TCP请求

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> Frame 2177: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{8A2BD733-6604-40FC-8041-F4CFBABB1E1E}, id 0
> Ethernet II, Src: Intel_70:f5:94 (4c:77:cb:70:f5:94), Dst: zte_c1:2a:f3 (78:90:a2:c1:2a:f3)
> Internet Protocol Version 4, Src: 192.168.1.17, Dst: 120.92.211.176
√ Transmission Control Protocol, Src Port: 1483, Dst Port: 23, Seq: 0, Len: 0
    Source Port: 1483
    Destination Port: 23
    [Stream index: 57]
    > [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 0 (relative sequence number)
    Sequence Number (raw): 3834764008
    [Next Sequence Number: 1 (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
    > Flags: 0x002 (SYN)
    Window: 64240
    [Calculated window size: 64240]
    Checksum: 0xa0fc [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    > Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted
    > [Timestamps]
```

2.服务器响应连接请求

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> Frame 2184: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{8A2BD733-6604-40FC-8041-F4CFBABB1E1E}, id 0
> Ethernet II, Src: zte_c1:2a:f3 (78:90:a2:c1:2a:f3), Dst: Intel_70:f5:94 (4c:77:cb:70:f5:94)
> Internet Protocol Version 4, Src: 120.92.211.176, Dst: 192.168.1.17
√ Transmission Control Protocol, Src Port: 23, Dst Port: 1483, Seq: 0, Ack: 1, Len: 0
    Source Port: 23
    Destination Port: 1483
    [Stream index: 57]
    > [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 0 (relative sequence number)
    Sequence Number (raw): 3500928766
    [Next Sequence Number: 1 (relative sequence number)]
    Acknowledgment Number: 1 (relative ack number)
    Acknowledgment number (raw): 3834764009
    1000 .... = Header Length: 32 bytes (8)
    > Flags: 0x012 (SYN, ACK)
    Window: 14400
    [Calculated window size: 14400]
    Checksum: 0xa404 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    > Options: (12 bytes), Maximum segment size, No-Operation (NOP), No-Operation (NOP), SACK permitted, No-Operation (NOP), Window scale
    > [Timestamps]
```

3.主机返回ACK完成三次握手成功建立连接

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> Frame 2185: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{8A2BD733-6604-40FC-8041-F4CFBABB1E1E}, id 0
> Ethernet II, Src: Intel_70:f5:94 (4c:77:cb:70:f5:94), Dst: zte_c1:2a:f3 (78:90:a2:c1:2a:f3)
> Internet Protocol Version 4, Src: 192.168.1.17, Dst: 120.92.211.176
✓ Transmission Control Protocol, Src Port: 1483, Dst Port: 23, Seq: 1, Ack: 1, Len: 0
  Source Port: 1483
  Destination Port: 23
  [Stream index: 57]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 3834764009
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 3500928767
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x010 (ACK)
  Window: 1029
  [Calculated window size: 263424]
  [Window size scaling factor: 256]
  Checksum: 0x1900 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > [Timestamps]

```

释放连接

3244	50.305647	120.92.211.176	192.168.1.17	TELNET	63 Telnet Data ...
3245	50.305647	120.92.211.176	192.168.1.17	TCP	54 23 → 1483 [FIN, ACK] Seq=10451 Ack=64 Win=14848 Len=0
3246	50.305720	192.168.1.17	120.92.211.176	TCP	54 1483 → 23 [ACK] Seq=64 Ack=10452 Win=263424 Len=0
3247	50.306202	192.168.1.17	120.92.211.176	TCP	54 1483 → 23 [FIN, ACK] Seq=64 Ack=10452 Win=263424 Len=0
3249	50.317325	120.92.211.176	192.168.1.17	TCP	54 23 → 1483 [ACK] Seq=10452 Ack=65 Win=14848 Len=0
4876	84.970188	192.168.1.17	120.92.211.176	TCP	74 Echo (ping) request id=0x0001 seq=2/512 ttl=128 (ren

1.TCP服务器发送一个FIN关闭服务器到客户端的数据传送 FIN=1

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> Frame 3245: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{8A2BD733-6604-40FC-8041-F4CFBABB1E1E}, id 0
> Ethernet II, Src: zte_c1:2a:f3 (78:90:a2:c1:2a:f3), Dst: Intel_70:f5:94 (4c:77:cb:70:f5:94)
> Internet Protocol Version 4, Src: 120.92.211.176, Dst: 192.168.1.17
✓ Transmission Control Protocol, Src Port: 23, Dst Port: 1483, Seq: 10451, Ack: 64, Len: 0
  Source Port: 23
  Destination Port: 1483
  [Stream index: 57]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 10451 (relative sequence number)
  Sequence Number (raw): 3500939217
  [Next Sequence Number: 10452 (relative sequence number)]
  Acknowledgment Number: 64 (relative ack number)
  Acknowledgment number (raw): 3834764072
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x011 (FIN, ACK)
  Window: 29
  [Calculated window size: 14848]
  [Window size scaling factor: 512]
  Checksum: 0xf3d5 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > [Timestamps]

```

2.客户端接收这个FIN发回一个ACK确认序号为收到的序号+1 ACK=1

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> Frame 3246: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{8A2BD733-6604-40FC-8041-F4CFBABB1E1E}, id 0
> Ethernet II, Src: Intel_70:f5:94 (4c:77:cb:70:f5:94), Dst: zte_c1:2a:f3 (78:90:a2:c1:2a:f3)
> Internet Protocol Version 4, Src: 192.168.1.17, Dst: 120.92.211.176
√ Transmission Control Protocol, Src Port: 1483, Dst Port: 23, Seq: 64, Ack: 10452, Len: 0
    Source Port: 1483
    Destination Port: 23
    [Stream index: 57]
> [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 64 (relative sequence number)
    Sequence Number (raw): 3834764072
    [Next Sequence Number: 64 (relative sequence number)]
    Acknowledgment Number: 10452 (relative ack number)
    Acknowledgment number (raw): 3500939218
    0101 .... = Header Length: 20 bytes (5)
> Flags: 0x010 (ACK)
    Window: 1029
    [Calculated window size: 263424]
    [Window size scaling factor: 256]
    Checksum: 0xefed [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
> [Timestamps]

```

3.服务器关闭客户端的连接，发送一个FIN给客户端 FIN=1,ACK=1

```

> Frame 3247: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{8A2BD733-6604-40FC-8041-F4CFBABB1E1E}, id 0
> Ethernet II, Src: Intel_70:f5:94 (4c:77:cb:70:f5:94), Dst: zte_c1:2a:f3 (78:90:a2:c1:2a:f3)
> Internet Protocol Version 4, Src: 192.168.1.17, Dst: 120.92.211.176
√ Transmission Control Protocol, Src Port: 1483, Dst Port: 23, Seq: 64, Ack: 10452, Len: 0
    Source Port: 1483
    Destination Port: 23
    [Stream index: 57]
> [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 64 (relative sequence number)
    Sequence Number (raw): 3834764072
    [Next Sequence Number: 65 (relative sequence number)]
    Acknowledgment Number: 10452 (relative ack number)
    Acknowledgment number (raw): 3500939218
    0101 .... = Header Length: 20 bytes (5)
> Flags: 0x011 (FIN, ACK)
    Window: 1029
    [Calculated window size: 263424]
    [Window size scaling factor: 256]
    Checksum: 0xefec [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
> [Timestamps]

```

4.客户端发回ACK报文确认，并将确认序号设置为收到序号+1 ACK=1

使用telnet时，客户端和服务端使用的IP地址和端口号分别是什么？

客户端：192.168.1.17:1483

服务器：120.92.211.176:23